

1. 线性. 满足 $ax_1(t) + bx_2(t) \rightarrow ay_1(t) + by_2(t)$
 时不变 $y(t) = \int_0^\infty \tau e^{-\tau} x(t-\tau) d\tau$ $y(t-t_0) = \int_0^\infty \tau e^{-\tau} x(t-t_0-\tau) d\tau = F[x(t-t_0)]$
 初始条件松弛条件下 (零状态), 用微分方程
 描述 LTI 系统 (系数全是常数) 是 LTI 系统.

因果. $h(t) = te^{-\tau} u(t)$, $t < 0$ 时 $h(t) = 0$

稳定. $\int_{-\infty}^{\infty} |h(\tau)| d\tau < \infty$, 即 $h(t)$ 绝对可积存在傅里叶
 变换. $\frac{1}{(2+j\omega)^2}$

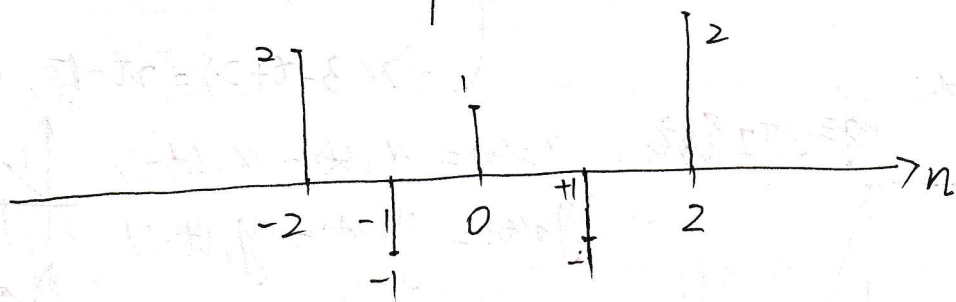
有记忆. $t \neq 0$ 时, $h(t) \neq 0$, 即 $h(t) \neq k\delta(t)$

可逆. 可找到 $h(t) * h(t) = \delta(t)$ 即 $H(\omega) = \frac{1}{H_1(\omega)}$
 $H_1(\omega) = (2+j\omega)^2$

$$\therefore y(t) = 4x(t) + 4x'(t) + x''(t)$$

2. $x(n) = \{2, 1, -2, 2, 2, 1, -1, -2, 1\}$, $-2 \leq n \leq 6$ $-2 \leq -2n^2 + 6 \leq 6 \Rightarrow -8 \leq -2n^2 \leq 0$
 $\Rightarrow 0 \leq n^2 \leq 4 \Rightarrow -2 \leq n \leq 2$

$x[-2n^2+6] = \{2, -1, 1, -1, 2\}$ 偶函数



3. $x(t) = 3 + \sin 4t + 2 \cos 6t$ $\omega_1 = 4, \omega_2 = 6 \Rightarrow \begin{cases} \omega_0 = 2 \\ T_0 = \pi \end{cases}$
 $-2B_2 = 1$ $2B_3 = 2$

$\therefore A_0 = 3, C_2 = -\frac{j}{2}, C_{-2} = \frac{j}{2}$

$C_3 = 1 = B_{-3}$

或 $\sin 4t = \frac{e^{j4t} - e^{-j4t}}{2j}$

$2 \cos 6t = e^{j6t} + e^{-j6t}$

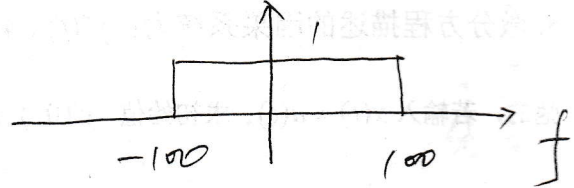
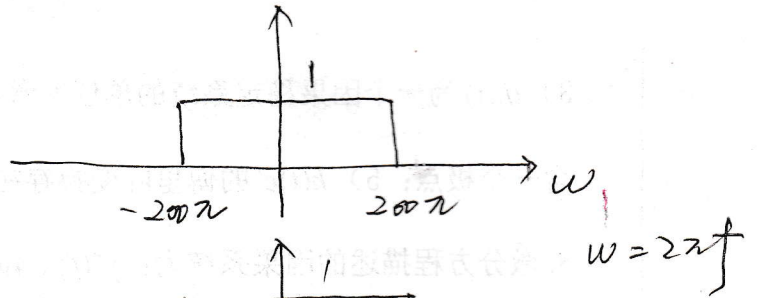
①

$\therefore C_2 = \frac{1}{2j}, C_{-2} = -\frac{1}{2j}$

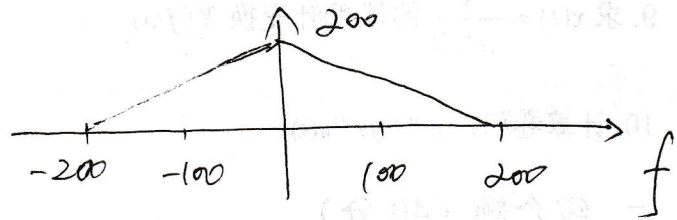
$\therefore C_3 = C_{-3} = 1$

$$4. f(t) = \left(\frac{\sin 200\pi t}{\pi t} \right)^2 * \cos 100\pi t = f_1^2(t) * f_2(t)$$

$$f_1(t) = \frac{\sin 200\pi t}{\pi t} \leftrightarrow F_1(\omega) =$$

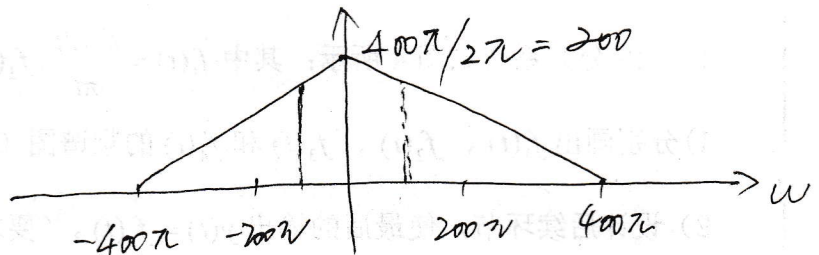


$$f_1^2(t) \leftrightarrow \frac{F_1(\omega) * F_1(\omega)}{2\pi} =$$



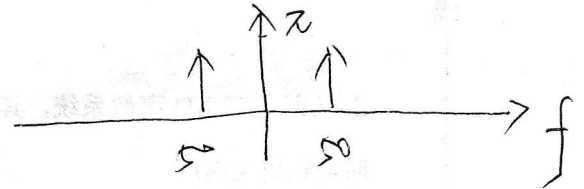
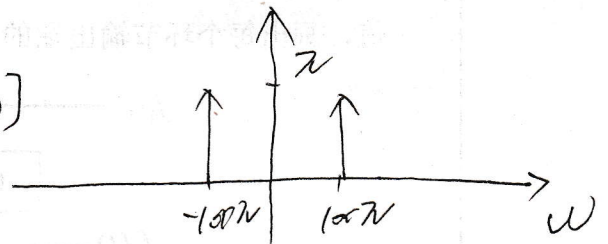
$$= F_1(f) * F_1(f)$$

$$= F_3(\omega)$$



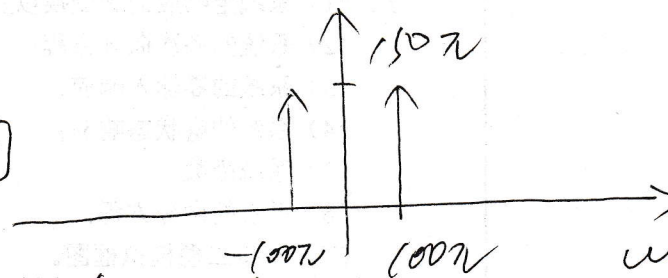
$$\cos 100\pi t \leftrightarrow \pi [\delta(\omega - 100\pi) + \delta(\omega + 100\pi)]$$

$$= F_2(\omega)$$



$$f(t) \leftrightarrow F_3(\omega) F_2(\omega) =$$

$$= 150\pi [\delta(\omega - 100\pi) + \delta(\omega + 100\pi)]$$

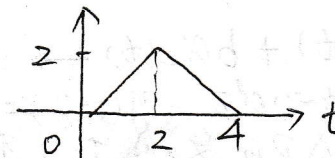


$$f(t) = 150 \cos 100\pi t, \quad f_0 = 50 \text{ Hz}, \quad \omega_0 = 100\pi \text{ rad/s}$$

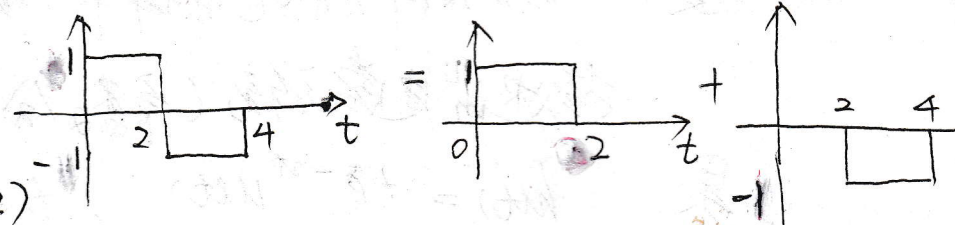
$$\therefore f_s \geq 2f_0 = 100 \text{ Hz}$$

$$\omega_s \geq 2\omega_0 = 200\pi \text{ rad/s}$$

5. $y_1(t) = x_1(t) * h(t) =$



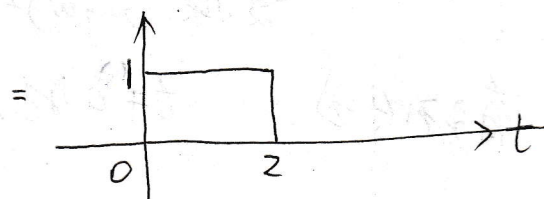
$y_1'(t) = x_1(t) * h'(t) =$



$= x_1(t) - x_1(t-2)$

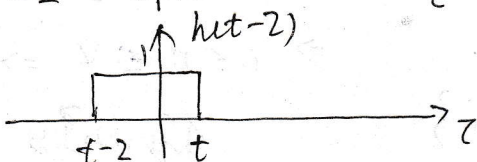
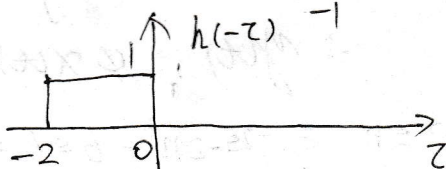
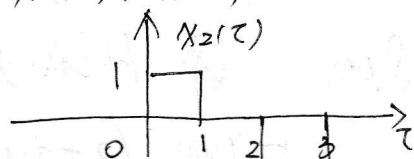
$= x_1(t) * \delta(t) + x_1(t) * [-\delta(t-2)]$

$\therefore h'(t) = \delta(t) - \delta(t-2) \Rightarrow h(t) = u(t) - u(t-2)$



$\therefore y_2(t) = x_2(t) * h(t)$

解法:



$y_2(t) = \begin{cases} \int_0^t 1 \cdot d\tau = t, & 0 \leq t \leq 1 \\ \end{cases}$

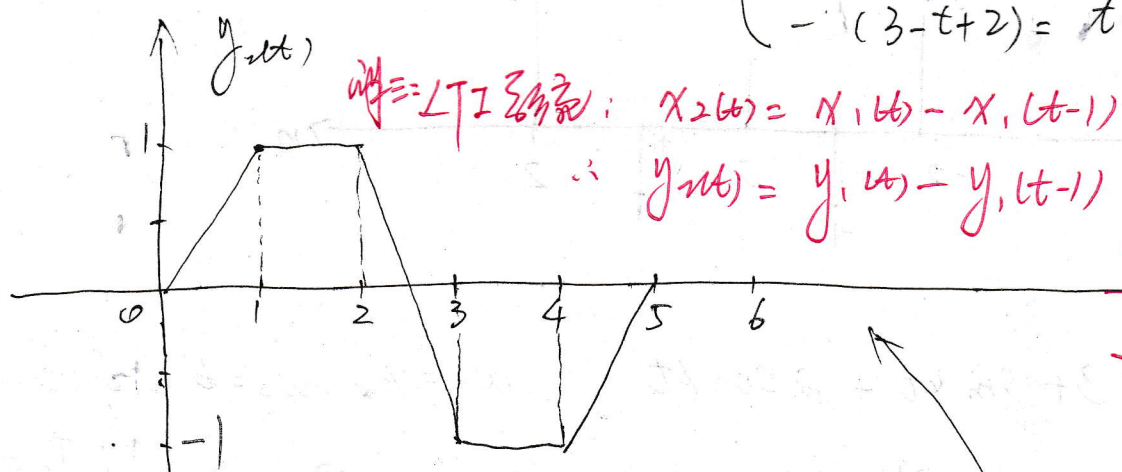
$\begin{cases} 1, & 1 < t < 2 \end{cases}$

$\begin{cases} (1-t+2) - (t-2) \end{cases}$

$\begin{cases} = 3-2t, & 2 \leq t \leq 3 \end{cases}$

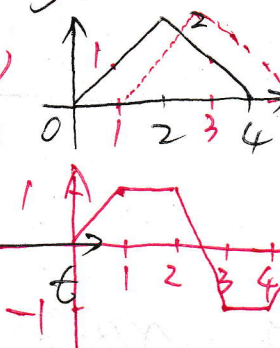
$\begin{cases} -1, & 3 < t < 4 \end{cases}$

$\begin{cases} -(3-t+2) = t-5, & 4 \leq t \leq 5 \end{cases}$

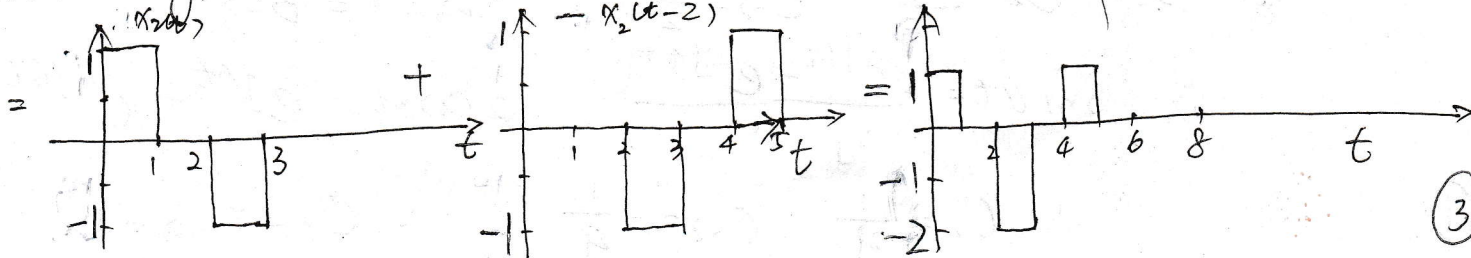


另解: LTI 系统: $x_2(t) = x_1(t) - x_1(t-1)$

$\therefore y_2(t) = y_1(t) - y_1(t-1)$



另解: $y_2'(t) = x_2(t) * h'(t) = x_2(t) * [\delta(t) - \delta(t-2)] = x_2(t) - x_2(t-2)$



2012 级 信号与系统 A 课程试卷 A 卷

开课学院: 信息与电子学院 试卷用途: ☐期中 ☐期末 ☐补考

考试形式: ☐开卷 ☐半开卷 ☐闭卷 考试日期: _____ 所需时间: 120 分钟

考试允许带: 笔, 尺子 入场

班级: _____ 学号: _____ 姓名: _____ 任课教师: _____

考生承诺: **“我确认本次考试是完全通过自己的努力完成的。”**

考生签名: _____

题序	一(简答题)	二(综合题)		总分
		1	2	100
满分	60	19	21	
得分				
评卷人				

备用数据:

$$6. \quad X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} = x[2]z^2 + x[1]z + x[0] + x[-1]z^{-1} + \dots \\ = z^2 + 2z + 3 - 2z^{-3} - z^{-4}$$

$$\therefore x[n] = \{1, 2, 3, 0, 0, -2, -1\}$$

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega \Rightarrow \int_{-\pi}^{\pi} X(e^{j\omega}) d\omega = 2\pi x[0] \\ = 6\pi$$

7. 因果稳定系统: $H(s)$ 根点在 s 域左半平面, 即 $H(s)$ 的 ROC 在最右一个根点以右, 其 ROC 必然包含虚轴。

单位冲激响应 $h(t)$ 应该是右边的. 即 $h(t)=0, t<0$ 时, 且 $h(t)$ 绝对可积

$$\int_{-\infty}^{\infty} |h(t)| dt < \infty \quad \text{即} \quad \int_0^{\infty} |h(t)| dt < \infty$$

(1) 错误. 如果 $h(t)$ 持续时间有限. 则 $h(t)$ 绝对可积. ROC 为全平面. 不存在极点.

(2) 错误. $\because H(s)$ 在复平面上不为 0.

$$H(s) = \int_{-\infty}^{\infty} h(t) e^{-st} dt \Rightarrow H(0) = \int_{-\infty}^{\infty} h(t) dt \neq 0.$$

(3) S域微分性质: $h(t) \leftrightarrow H(s) \quad \text{ROC} = R$ 收敛域不变.

$$t h(t) \leftrightarrow -\frac{dH(s)}{ds} \quad \text{ROC} = R$$

$\therefore t h(t)$ 仍为一个因果稳定系统的单位冲激响应. $\therefore$ 正确.

(4) ~~不正确~~. 根据单边信号拉氏变换的时域微分性质.

$$\frac{dh(t)}{dt} \leftrightarrow sH(s) - h(0^-) \quad \text{存在常数 } h(0^-), \text{ 通分后可能}$$

~~发生零极点抵消.~~

$$\because t < 0 \text{ 时 } h(t) = 0 \therefore \frac{dh(t)}{dt} = sH(s)$$

(5) 不正确. 根据 S域平移性质. $h(t)e^t \leftrightarrow H(s-1)$

收敛域变为 $R+1$. 向右移 1, 可能不包含虚轴. 则可能不存在傅里叶变换. 可能存在其他极点.

(6) 错误 $H(s) = H(-s)$, 存在对称极点. 已有一极点为 $s = -2$, 则另一极点为 $s = 2$. 不满足因果性. $\therefore$ 错误.

8. 初始状态为 0, 初始状态为 0. $s^2 Y(s) + 4s Y(s) + 5 Y(s)$

$$= \frac{1}{2} s X(s) + X(s)$$

$$Y(s) = \frac{\frac{1}{2}s + 1}{s^2 + 4s + 5} X(s)$$

$$x(t) = u(t) \Rightarrow X(s) = \frac{1}{s}$$

(5)

$$\therefore Y(s) = \frac{\frac{1}{2}s+1}{s(s^2+4s+5)}$$

初值定理 $y(0^+) = \lim_{s \rightarrow \infty} sX(s) = \lim_{s \rightarrow \infty} \frac{\frac{1}{2}s+1}{s^2+4s+5} = 0$

$$L[y'(t)] = sY(s) - y(0^-) = \frac{\frac{1}{2}s+1}{s^2+4s+5}$$

$$y'(0^+) = \lim_{s \rightarrow \infty} s L[y'(t)] = \lim_{s \rightarrow \infty} \frac{\frac{1}{2}s^2+s}{s^2+4s+5} = \lim_{s \rightarrow \infty} \frac{1}{2} + \frac{-s-\frac{5}{2}}{s^2+4s+5}$$

$$= \frac{1}{2}$$

微分方程法: $\begin{cases} y''(t) + 4y'(t) + 5y(t) = \frac{1}{2}\delta(t) + ut \\ y'(0^-) = y(0^-) = 0 \end{cases}$ 求 $y'(0^+)$ 和 $y(0^+)$

函数匹配法. 右端有 $\delta(t)$. 则左端 $y''(t)$ 中有 $\delta(t)$ 项. $y'(t)$ 中有 ut 项.

对方程左右两边取 $0^- \sim 0^+$ 的定积分. 有.

$$\int_{0^-}^{0^+} y''(t) dt + 4 \int_{0^-}^{0^+} y'(t) dt + 5 \int_{0^-}^{0^+} y(t) dt = \frac{1}{2} \int_{0^-}^{0^+} \delta(t) dt$$

$$\text{则 } y'(0^+) - y'(0^-) + 4y(0^+) - 4y(0^-) = \frac{1}{2} + \int_{0^-}^{0^+} ut dt$$

$y(t)$ 中有 ut 项. 在 $t=0$ 处不连续. $y(t)$ 在 $t=0$ 处连续.

$$\therefore y(0^+) = 0, \quad y'(0^+) = \frac{1}{2}$$

9. $\frac{1}{2+j\omega} \longleftrightarrow e^{-2t} u(t)$

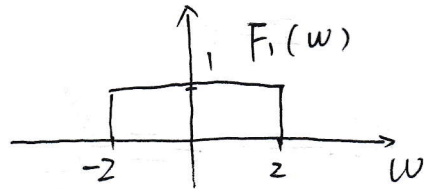
对称性: $\frac{1}{2+jt} \longleftrightarrow 2\pi e^{2w} u(-w)$

10. $e^{st} \longrightarrow H(s)e^{st} \quad H(s) = L[te^{-4t} u(t)] = \frac{1}{(s+4)^2}$

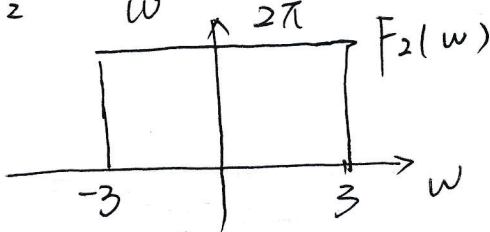
$\therefore \mathcal{B} e^{-3t} * [te^{-4t} u(t)] = \left. \frac{1}{(s+4)^2} \right|_{s=-3} \cdot e^{-3t} = e^{-3t} \quad (6)$

证: $A(t) * [B(t) \cdot C(t)] \neq A(t) * B(t) \cdot C(t)$

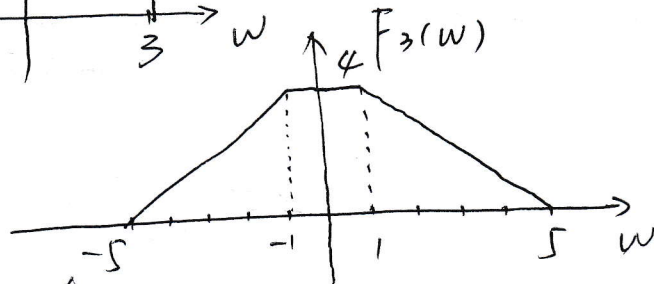
$$f_1(t) = \frac{\sin 2t}{\pi t}$$



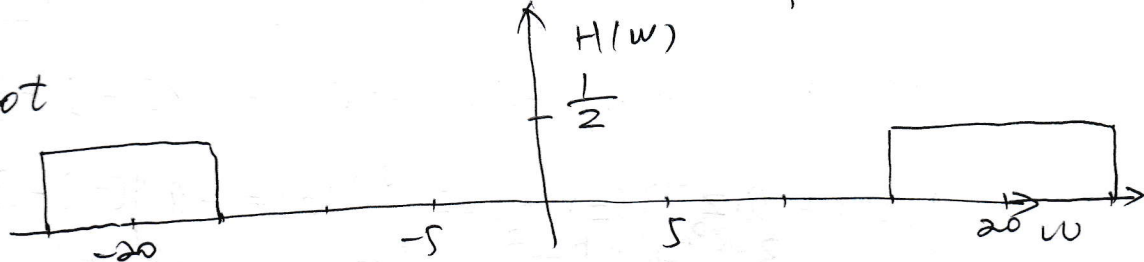
$$f_2(t) = \frac{2 \sin 3t}{t} = 2\pi \cdot \frac{\sin 3t}{\pi t}$$



$$f_3(t) = f_1(t) \cdot f_2(t) \Rightarrow F_3(w) = \frac{F_1(w) * F_2(w)}{2\pi}$$

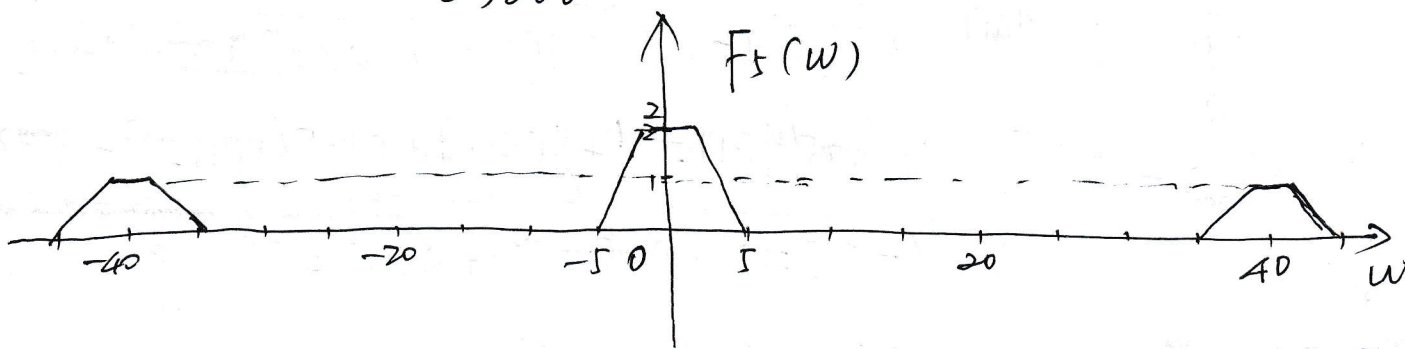
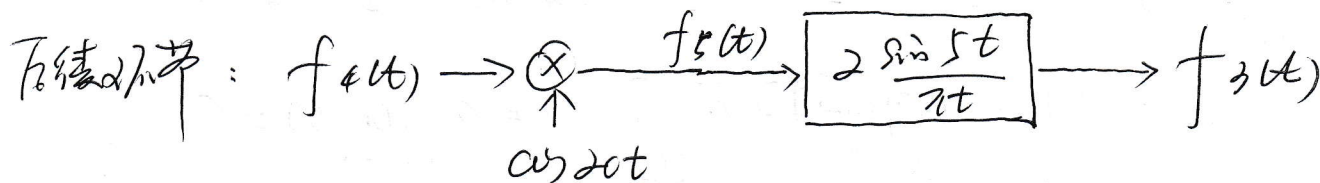
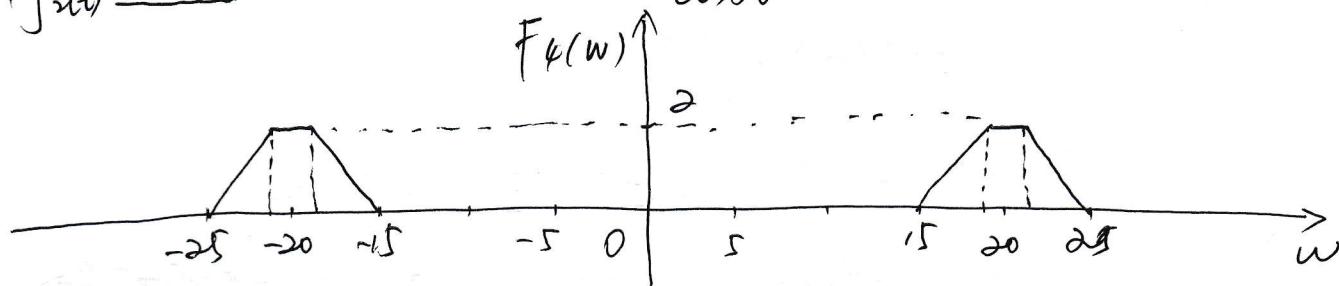
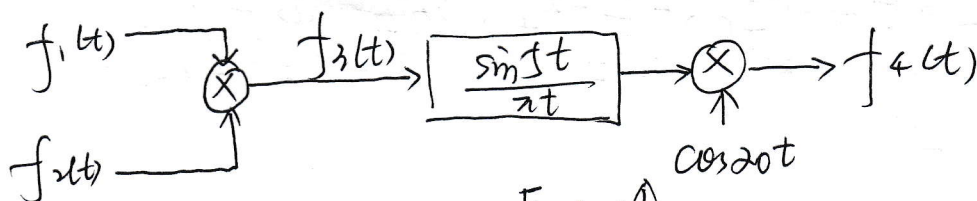


$$h(t) = \frac{\sin 5t}{\pi t} \cdot \cos 20t$$



$\therefore f_4(t) = 0$

2) 由于第(1)问有京问题. 将题目修改如下:



(7)

$$2. 1) (0.5)^n u[n] \leftrightarrow \frac{1}{1 - 0.5z^{-1}} = \frac{z}{z - 0.5} \quad |z| > 0.5$$

$$n(0.5)^n u[n-1] = n(0.5)^n u[n] \leftrightarrow -z \frac{d}{dz} \frac{z}{z - 0.5} = \frac{0.5z}{(z - 0.5)^2}$$

$$(-0.5)^n u[n] \leftrightarrow \frac{1}{1 + 0.5z^{-1}} = \frac{z}{z + 0.5} \quad |z| > 0.5$$

$$\therefore Y(z) = \frac{4z}{z - 0.5} - \frac{0.5z}{(z - 0.5)^2} - \frac{z}{z + 0.5} \quad (|z| > 0.5)$$

$$a_0 y[n] + a_1 y[n-1] + a_2 y[n-2] = b_0 x[n] + b_1 x[n-1] + b_2 x[n-2]$$

$$a_0 Y(z) + a_1 z^{-1} Y(z) + a_2 z^{-2} Y(z) = -a_1 y[-1] - a_2 z^{-1} y[-1] - a_2 y[-2] + b_0 X(z) + b_1 X(z) z^{-1} + b_2 z^{-2} X(z)$$

$$Y(z) = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{a_0 + a_1 z^{-1} + a_2 z^{-2}} X(z) - \frac{a_1 y[-1] + a_2 y[-2] + a_2 z^{-1} y[-1]}{a_0 + a_1 z^{-1} + a_2 z^{-2}}$$

(2)

$$= \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{a_0 + a_1 z^{-1} + a_2 z^{-2}} \cdot \frac{1}{1 - 0.5z^{-1}} - \frac{8a_1 + 2a_2 + 8a_2 z^{-1}}{a_0 + a_1 z^{-1} + a_2 z^{-2}}$$

$$\begin{cases} y[n] \leftrightarrow Y(z) \\ u[n] y[n-1] \leftrightarrow z^{-1} Y(z) + y[-1] \\ u[n] y[n-2] \leftrightarrow z^{-2} Y(z) + z^{-1} y[-1] + y[-2] \end{cases}$$

$$\therefore a_0 + a_1 z^{-1} + a_2 z^{-2} = (1 - 0.5z^{-1})(1 + 0.5z^{-1}) = 1 - 0.25z^{-2} \Rightarrow \begin{cases} a_0 = 1 \\ a_1 = 0 \\ a_2 = -0.25 \end{cases}$$

齐次差分方程 $y[n] - 0.25y[n-2] = 0$

$$(3) \text{ 零输入响应: } Y_0(z) = \frac{0.5 + 2z^{-1}}{1 - 0.25z^{-2}} = \frac{A}{1 + 0.5z^{-1}} + \frac{B}{1 - 0.5z^{-1}}$$

$$\Rightarrow \begin{cases} A + B = 0.5 \\ -0.5A + 0.5B = 2 \end{cases} \Rightarrow \begin{cases} A = -\frac{7}{4} \\ B = \frac{9}{4} \end{cases} \Rightarrow y_0[n] = -\frac{7}{4}(-0.5)^n u[n] + \frac{9}{4}(0.5)^n u[n]$$

(8)

$$y_x[n] = y[n] - y_0[n] = \frac{7}{4} (0.5)^n u[n] - n(0.5)^n u[n-1] + \frac{3}{4} (-0.5)^n u[n]$$

$$(5) \quad Z[y_x[n]] = H(z)X(z) = \frac{\frac{7}{4}}{1-0.5z^{-1}} - \frac{0.5z^{-1}}{(1-0.5z^{-1})^2} + \frac{\frac{3}{4}}{1+0.5z^{-1}}$$

$$= H(z) \cdot \frac{1}{1-0.5z^{-1}}$$

$$\therefore H(z) = \frac{7}{4} - \frac{0.5z^{-1}}{1-0.5z^{-1}} + \frac{\frac{3}{4}(1-0.5z^{-1})}{1+0.5z^{-1}}$$

$$= \frac{\frac{7}{4}(1-0.5z^{-2}) - 0.5z^{-1}(1+0.5z^{-1}) + \frac{3}{4}(1-0.5z^{-1})^2}{1-0.25z^{-2}}$$

$$= \frac{2.5 - 1.25z^{-1} - 0.5z^{-2}}{1-0.25z^{-2}}$$

$$(b) \quad b_0 = 2.5, \quad b_1 = -1.25, \quad b_2 = -0.5$$

$$\therefore \text{差分方程: } y[n] - 0.25y[n-2] = 2.5x[n] - 1.25x[n-1] - 0.5x[n-2]$$

